Legacy record: the uncorrected seed-3–9 models and superseded results

Working report (supplement)

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Abstract

This supplement documents a set of networks (seeds 3–9, sizes 128 and 256) trained under an unintended regularisation coefficient, and the analysis outputs derived from them, which have been superseded by re-trained ("corrected") models. It exists for provenance: the main report analyses the corrected 256-unit seeds 3–9 (plus released seeds 0–2) and uses only the released versions at the other network sizes. No conclusions are drawn here.

1 The regularisation-coefficient issue

A training script specified the L1 activity (ℓ_{1h}) and L1 weight (ℓ_{1w}) penalties by an exponent e that was mapped to a coefficient by 10^e with no guard. The value intended to disable the penalty, $e = 0$, produced $10^0 = 1$ rather than 0; the networks were therefore trained with an L1 penalty of magnitude 1 on the mean absolute activity and on the weights (versus the intended 0; the L2 coefficients, 10^{-6}, were unaffected). The other training scripts in the codebase guard this mapping as $\min(-e, 10^e)$ (so $e = 0 \mapsto 0$, $e = -6 \mapsto 10^{-6}$); the affected script did not. It was corrected to use the same guard and seeds 3–9 were re-trained.

2 Characteristics of the uncorrected models (256-unit)

quantity	uncorrected seeds 3–9	corrected seeds 3–9
ℓ_{1h}, ℓ_{1w} coefficients	1, 1	0, 0
L2 coefficients	10^{-6}	10^{-6}
active units (variance $> 10^{-3}$)	≈ 71	212–220
operating scale (20% of active)	14–15	42–44
mean accuracy (worst task)	0.91–0.94	0.94–0.96
mean accuracy (across tasks)	0.92–0.96	≈ 0.975

The uncorrected models had roughly one third as many active units, and correspondingly a smaller operating group number, because the large L1 activity penalty drove most units to zero activity.

3 Superseded analysis outputs

The following outputs were computed on the uncorrected models and are superseded by the corresponding corrected-model results in the main report. The direction of every effect was preserved

after correction; the magnitudes changed. Representative examples (256-unit, $k = 20$ or operating scale as applicable):

quantity	uncorrected	corrected	main-report section
motif-direction containment score (median)	+2.52	+7.51	containment
quotient prediction A_{cond} (median)	(n/a) †	0.816	quotient reduced model
mode-concentration PR conn vs. random (10/10)	preserved	preserved	structural bottleneck
rotation-readout ring > matched	preserved	104/109	rotation

† The quotient-prediction, eigenvalue-matched, taxonomy, and reuse analyses were recomputed from scratch on the corrected models; per-seed uncorrected values were not carried forward except where a direct before/after comparison was recorded (e.g. the containment score).

4 Storage

The uncorrected checkpoints and their derived structural outputs (activity clusters, signed-WL sweeps for 256-unit seeds 3–9) were moved to a backup directory (`results/_buggy_l1_backup_20260726/`) prior to re-running the pipeline on the corrected models. The uncorrected fixed-point trees were overwritten in place by the corrected-model re-run.

5 Other network sizes

Networks at 128, 512 and 1024 units were also trained for seeds 3–9 under the same uncorrected setting; the 128-unit set was subsequently re-trained. In the main report the non-256 sizes are represented only by the released seeds 0–2; the retrained non-256 seeds are not analysed there.